

Longitudinal Multimodal Machine Learning for Dementia Research: Methods, Gaps and Opportunities

Sara Calhas^{1,2}, Petroula Proitsi^{1,3}, Natalia Vilor-Tejedor^{2,4}

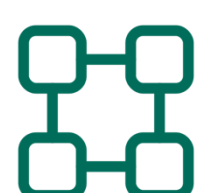
¹ Centre for Preventive Neurology, Queen Mary University of London, UK. ² Barcelonaβeta Brain Research Center (BBRC), Pasqual Maragall Foundation, Barcelona, Spain.

³ Department of Basic and Clinical Neuroscience, King's College London, UK. ⁴ Institute for Risk Assessment Sciences (IRAS), Utrecht University, Utrecht, Netherlands.

1. BACKGROUND



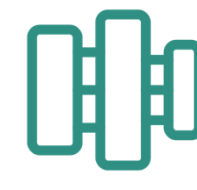
Dementia has a long preclinical and prodromal stage with subtle, non-linear trajectories.



Multimodal longitudinal data are rapidly expanding (imaging, clinical data, omics, genetics, digital).



Irregular sampling and missing observations are inherent to longitudinal dementia studies.



Machine learning (ML) and deep learning may offer new opportunities to tackle these challenges.

Scoping review to identify advanced ML methods for multimodal longitudinal modelling.

2. LLM-ASSISTED SCREENING

Literature search (2020-2025)

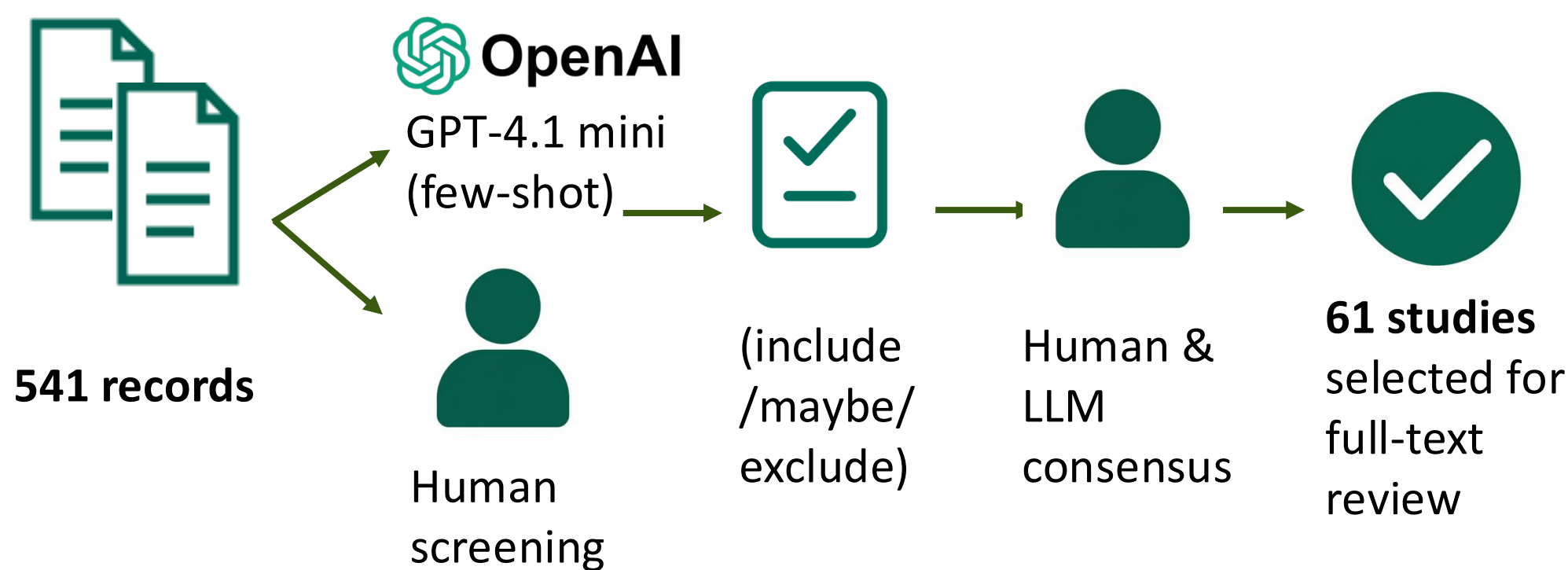


541 records after duplicates removed

LLM-assisted title and abstract screening

Pipeline using OpenAI API (GPT-4.1 mini, few-shot prompting)

A. Workflow overview



B. Performance vs manual screening

	Recall	Precision	Specificity	
Keep class	0.81	0.63	0.94	• High recall for Keep • High precision for Exclude
Exclude class	0.55	0.91	0.80	
Maybe class	0.55	0.14	0.65	

C. Efficiency

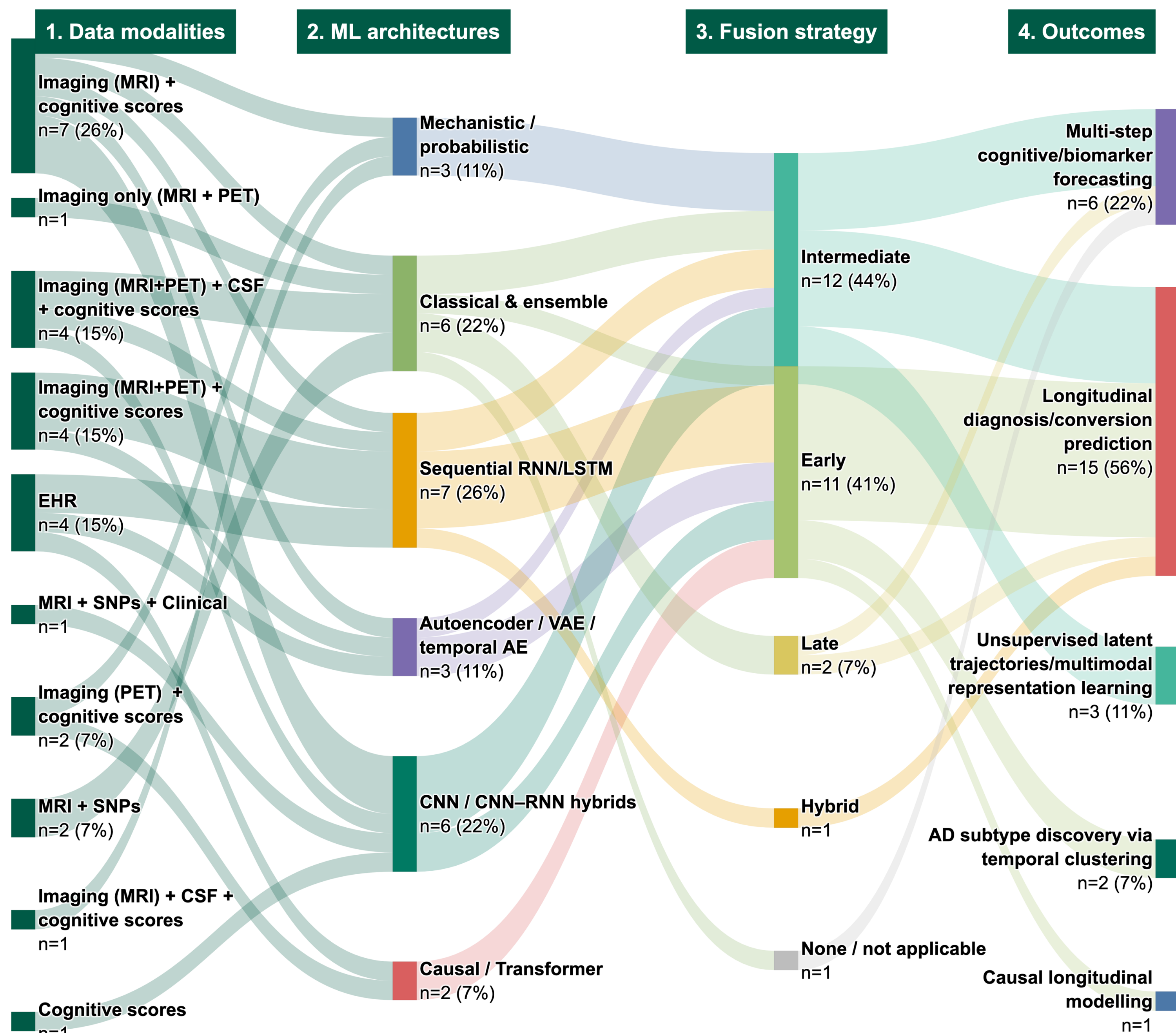
LLM-assisted screening reduced screening time from ~4 days to ~15 minutes while maintaining high recall for relevant studies.

Key takeaway

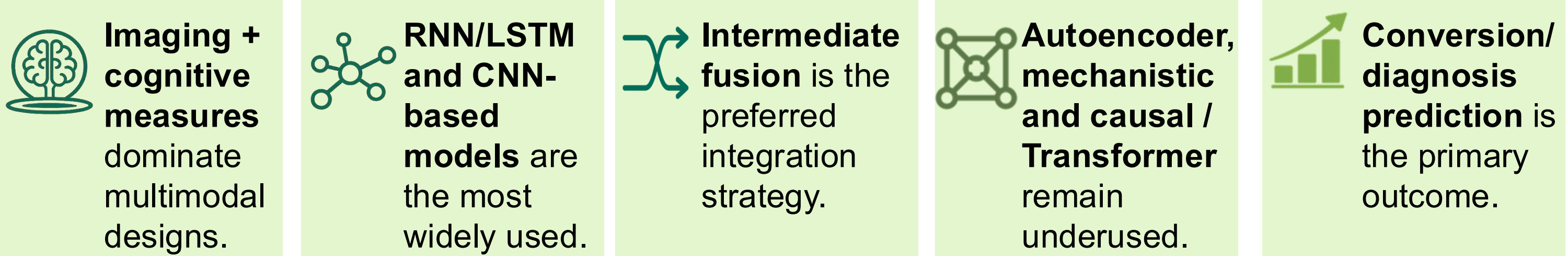
- LLM robust in identifying relevant studies and excluding irrelevant ones but struggled with borderline and ambiguous abstracts.
- Further tuning required for Maybe class.
- Proof of concept suggested **potential to accelerate** title and abstract screening.

3. LONGITUDINAL MACHINE LEARNING IN DEMENTIA

Current ML landscape (27 studies)



Key insights from the landscape



TAKE-HOME MESSAGE



Longitudinal multimodal ML is rapidly evolving but remains dominated by conventional architectures.

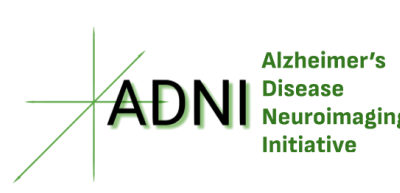


Few studies explicitly model latent disease trajectories, irregular sampling, and missingness together.



Representation learning, continuous-time, and causal approaches are key opportunities for next-generation dementia models.

4. KEY GAPS



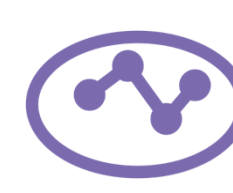
Datasets

21/27 studies used ADNI.



Sample size

23/27 studies included ≤2,000 participants.



Latent disease trajectories

Few studies learned shared latent disease trajectories.



External validation

Only 4 studies externally validated models.



Missing data and irregular intervals

Only 4 studies jointly modelled both.

5. FUTURE OPPORTUNITIES



Latent representation learning

Learning biologically meaningful latent disease trajectories underlying progression.



Multimodal temporal integration

Joint modelling of heterogeneous longitudinal data collected at different follow-up schedules while explicitly accounting for irregular sampling and missing observations.



Causal and mechanistic approaches

Moving beyond prediction by modelling biological disease mechanisms, hidden confounding and counterfactual intervention effects.

GET IN TOUCH

Sara Calhas
s.calhas@qmul.ac.uk



Wellcome Trust PhD Programme in Science (218584/Z/19/Z)

